Lab Shift Roster — user guide

A secure and intelligent workforce planning tool for diagnostic laboratories. Always a draft for your review.

This guide covers everything from filling in the workbook to exporting a finished rota. Version 1.1 Beta.

Use the tool at https://labshiftroster.com/app.

CONTENTS

1. What it does
2. Your data
3. Quick start
4. Choosing a workbook
5. Roster Details
6. Rules
7. Staff
8. Week Patterns
9. Competencies
10. Shifts
11. Shift Requirements
12. Benches
13. Leave & Leave Types
14. Generating a roster
15. Reading the dashboard
16. Working through issues
17. Roster preview
18. Contracted hours
19. Resilience
20. Adjusting by hand
21. Exporting
22. How it decides
23. If something goes wrong
24. What it does not do

1. What it does

Lab Shift Roster builds a draft rota for a diagnostic laboratory from a spreadsheet of your workforce, then tells you where that rota is thin.

The question it exists to answer is: **do I have the right people, with the right competencies, in the right laboratory areas, at the right time?**

It understands the things that make laboratory rostering awkward:

- Contracted hours and FTE, so a ten-hour night is not treated as equivalent to a seven-and-a-half hour day
- Part-time and alternating working patterns
- Competency as a status with an expiry date, not a tick-box
- Grade, seniority and competence as three separate things
- Sections that each need their own person standing at them
- Rest intervals measured from real shift times, including across midnight

> **Every roster it produces is a draft for you to review.** It supports your decisions; it does not make them. It is not artificial intelligence — it is a rules engine that fills the shifts you define and tells you when it cannot.

2. Your data

> **Your workbook never leaves your device.** Lab Shift Roster runs entirely inside your browser. Your workbook and the roster it produces are not uploaded to Optymum SS, there is no account, and no roster is stored on any external database. Once the page has loaded you can disconnect from the network and it will still work.

What that means in practice:

- The workbook you choose is read by the page itself, not sent anywhere
- The rota you download is written on your own computer
- Closing the tab discards everything; nothing persists

Usage analytics

The web page itself does use Google Analytics, so limited information about how Lab Shift Roster is used is transmitted: page views, and three events recording that a draft roster was generated, a staff rota was exported or a manager report was exported. Those events are names alone.

No staff, roster, competency, leave, hours, organisation or workbook information is included — not staff names or identifiers, not your file name, not any figure from your workbook, and no error text. Advertising features, Google Signals and ad personalisation are switched off. The distinction worth holding on to: your workforce data stays local; ordinary website analytics does not.

No absolute security claims are made about your own device or network. Please include only the workforce information the tool needs. Do not add unnecessary personal details, and **never add patient-identifiable information** — none is required.

3. Quick start

If you want to see what it produces before entering any real data, open the application and press **Try the balanced example**. That generates a rota from a fictional department in a few seconds.

For your own laboratory:

1. **Download a workbook** — the blank one, or an example to work from
2. **Fill it in** — staff, competencies, shifts, requirements, leave
3. **Upload it** — choose the file, nothing is transmitted
4. **Generate a draft roster**
5. **Review the issues** — start with Critical
6. **Export** — a staff rota to circulate, a manager report to keep

Expect the first attempt to raise issues. That is the tool doing its job: it is showing you where the plan is thin, not failing.

4. Choosing a workbook

Three downloads are offered, and they serve different purposes.

DOWNLOAD	CONTAINS	USE IT WHEN
Blank workbook	All the headings, guidance and dropdown lists, and nothing else: no staff, competencies or leave, and no shifts, requirements, benches or absence types either. Only the roster dates and your usual Rules settings are filled in, so check them.	You are setting up your own laboratory
Balanced example	A well-staffed fictional department of twenty-one, running eight benches. Produces full coverage with nothing to review: no critical items, no review items, only passed checks.	You want to see what a healthy result looks like
Challenging example	A fictional department with deliberate gaps: thin morphology cover, an expired competency, part-time constraints and leave.	You want to see what the tool detects

> Every name, competency and date in both examples is invented. Neither contains real staff information, and the challenging example is labelled as deliberately difficult so nobody mistakes it for a typical result.

The workbook has eleven sheets. The next sections cover each one. You do not need to complete every column — most have sensible behaviour when left blank — and the guidance below says which ones matter.

5. Roster Details

Who the rota is for and what period it covers. These appear on the exported workbook, so they are worth filling in properly if the rota will be circulated.

SETTING	NOTES
Rota name	Defaults to `Laboratory Staff Rota`
Organisation	Appears in the export heading
Department	For example *Blood Sciences*
Site	Useful where a trust runs several laboratories
Prepared by	Recorded in the manager report
Roster period start	Required. Enter as `dd/mm/yyyy`
Roster period end	Required. Inclusive
Workbook version	Leave alone. Lets the tool recognise an older workbook

A month is the usual period. Longer works, but produces a very wide spreadsheet — you will be warned beyond a year.

6. Rules

Your laboratory's own operating rules. Every one is a setting you control.

> **These are planning settings, not compliance determinations.** The rest interval in particular is whatever your organisation has agreed. Lab Shift Roster reports a *configured rest rule conflict*; it never claims a shift is unlawful.

RULE	DEFAULT	WHAT IT DOES
Minimum rest hours between shifts	`11`	Measured from real finish and start times. A late ending 21:00 followed by an early starting 07:00 is ten hours and will be prevented
Maximum consecutive days	`6`	Longest unbroken run of working days. Can be overridden per person
Maximum consecutive nights	`4`	Longest unbroken run of nights
Night block length	`3`	Nights are rostered in blocks rather than scattered
Recovery days after nights	`2`	Days off following a block of nights
Hours tolerance (%)	`10`	How far from contracted hours before somebody is flagged
Competency expiry warnings (days)	`30, 60, 90`	Warn when a competency expires within these periods
Max simultaneous bench assignments	`1`	How many sections one person may cover at once. Leave at 1
Cross cover allowed	`N`	Reserved for a future release
Weekend days	`Saturday, Sunday`	Which days count as the weekend
Share nights evenly	`Y`	Spread nights across everyone eligible. See the note below
Section rotation warning (days)	`56`	Warn if somebody has not worked a section for this long
Alternative roster number	`42`	Changes which of several valid drafts you get. You never need to edit this — use the button instead

A note on sharing nights

Sharing nights evenly is on by default, because concentrating them on a few people while eligible colleagues do none is not acceptable in practice. It has a genuine cost: a block of nights also books the recovery days after it, so spreading nights across more people fragments cover for the day service. On the balanced example, turning sharing on moves requirement compliance from 100% to about 95%. Set it to `N` only if that last few per cent of cover matters more to you than sharing the unpopular work.

There is deliberately **no senior band threshold**. Seniority is recorded per person on the Staff sheet, and a shift that needs a particular grade uses `Min Band` on the Requirements sheet. A Band 7 is not automatically senior, and a senior is not automatically competent in everything.

7. Staff

One row per person. This is the most important sheet, and the one worth taking time over.

Identity and role

COLUMN	NOTES
Staff ID	Required and must be unique. Every other sheet refers to it. Anything consistent works — `S01`, initials, a payroll number
Name	Required. Shown on the rota
Initials	Optional. These are the initials used in the section allocation rows, and they now appear in their own column on the rota so a reader can tell whose they are. Leave it blank and they are worked out from the name and kept unique; fill it in if your laboratory already uses its own
Job Title	Shown in the manager report
Band	NHS band or local grade. `8a` sorts above `8`, which sorts above `7`
Registered BMS	`Y` / `N`. Used by the *Min Registered BMS* requirement
Senior	`Y` / `N`. Recorded explicitly, never inferred from band
Shift Coordinator	`Y` / `N`. Checked separately from seniority
Trainee	`Y` / `N`. Counted against *Max Trainees*

Hours

COLUMN	NOTES
Contracted Weekly Hours	Drives how much work each person is given. This is the main workload control
FTE	Used only if weekly hours are blank, applied to a 37.5-hour week
Working Pattern	Free text label such as *Part time*. Descriptive only
Max Hours This Period	Hard ceiling for the whole period. Blank means no ceiling
Max Weekly Hours	Hard ceiling per week

When they are available

COLUMN	NOTES
Mon – Sun	`Y` / `N` per day: the days this person is available. **Leave all seven blank for somebody fully flexible.** These are hard constraints — a blank Saturday means they can never be rostered on a Saturday

COLUMN	NOTES
Pattern Cycle (weeks)	Set to `2` for an alternating fortnight, then use the Week Patterns sheet
Earliest Start / Latest Finish	Times such as `09:00` and `16:00`. A shift that falls outside them is never offered
Max Days Per Week	Personal cap
Max Consecutive Days	Overrides the organisational rule for this person
Works Nights / Works Weekends	`Y` / `N` eligibility
Max Nights / Max Weekends	Caps for the period
Group	Splits the rota into labelled blocks, such as *Main* and *Bank*
Restrictions / Notes	Your own reminders. Restrictions appear in the manager report only, never on the staff rota

> **The commonest setup mistake.** The Mon–Sun columns are *availability*, not a record of contracted days. If you mark somebody as working Monday to Friday only, they can never be rostered at a weekend however *Works Weekends* is set. If they are flexible, leave those columns blank.

8. Week Patterns

Only needed for staff who work alternating weeks. Set *Pattern Cycle (weeks)* on the Staff sheet first, then add one row here per person per week.

STAFF ID	WEEK	MON	TUE	WED	THU	FRI	SAT	SUN
S06	1	Y	Y	Y	N	N	N	N
S06	2	N	N	Y	Y	Y	N	N

Weeks are counted from the Monday of the week the roster period starts, so a period beginning midweek does not shift the pattern. If a cycle is declared but a week is left out, week 1 is used for it rather than making that person unavailable.

9. Competencies

One row per person per bench (a bench is also called a discipline: the two words mean the same thing). This is what lets the tool answer whether a bench can actually be staffed.

COLUMN	NOTES
Staff ID	Must match the Staff sheet
Name	For your readability; the ID is what is used
Bench or Discipline	The bench, written as on the Benches sheet: either its short **Bench** name (`GRP` , `FBC` , `COAG` , `MORPH`) or its full **Discipline** name (*Grouping*, *Full Blood Count*). Either works, and they are treated as the same. An older workbook whose column is headed *Discipline* still reads correctly
Competency	The name of the specific competency, such as *Crossmatch and electronic issue*
Status	See the table below
Date Achieved / Review Date	Your records
Expiry Date	Past this date the competency stops counting, whatever the status says
Trainer / Assessor	`Y` / `N` . Used by *Min Trainers*
Result Authoriser	`Y` / `N` . Checked separately from competence

What each status means

STATUS	COUNTS AS COVER?	MEANING
Not Trained	No	Has not started
In Training	No	Learning, and needs somebody with them
Supervised	No	Working towards sign-off, still supervised
Competent	Yes	Can work the discipline unsupervised
Trainer	Yes	Competent, and can train others
Assessor	Yes	Competent, and can assess others
Expired	No	Lapsed and needs reassessment

> Only **Competent**, **Trainer** and **Assessor** satisfy a requirement. Somebody *In Training* or *Supervised* needs a colleague present, so counting them as cover would overstate what the laboratory can actually deliver. An expired record never counts, even if the status column still says *Competent*.

Somebody may hold several records in one discipline. The strongest current one governs, so an old *In Training* row does not hide a later *Competent* one.

Authorisation is checked **independently** of competence. A result authoriser must also be currently competent in that discipline — the flag alone is not enough.

10. Shifts

The shifts your laboratory runs, defined by real clock times.

COLUMN	NOTES
Code	Short label shown in every cell of the rota, such as `C`
Name	Full name, such as *Core*
Start / End	24-hour times. Length is worked out from these, so the hours can never disagree with the times
Days	`All`, `Weekday` or `Weekend`
Active	`Y` or `N`. Set it to `N` for a shift your laboratory does not run. Leave it blank and the shift runs, so older workbooks are unaffected
Night Shift	`Y` brings in the night block, recovery days and consecutive-night rules
Colour / Font Colour	Six-digit hex. The cells show the actual colour and the Preview column shows how the shift will look

The starting configuration is:

CODE	NAME	TIMES	HOURS	DAYS	NIGHT
E	Early	07:00–15:00	8	Weekday	No
C	Core	09:00–17:30	8.5	Weekday	No
L	Late	13:00–21:00	8	Weekday	No
N	Night	21:00–07:00	10	All	Yes
W	Weekend Day	09:00–17:30	8.5	Weekend	No

A shift crossing midnight is handled correctly: `21:00` to `07:00` is ten hours, not a negative number. A shift that starts and finishes at the same time is rejected as a data-entry error.

Shifts your laboratory does not run

The blank workbook has no shifts: add one row for each shift your laboratory runs. The five above are what the example laboratories use, a starting point rather than a prescription. Plenty of laboratories have no early shift, or run a twilight instead of a late, or work weekends on the same pattern as weekdays.

Set Active to `N` for any shift you do not run. An inactive shift is never rostered, and any requirement or bench that mentions it is quietly ignored for as long as it is switched off — so you do not have to go and unpick the Shift Requirements and Benches sheets to remove one shift.

To add a shift, add a row. Give it a code that nothing else uses, then refer to that code from Shift Requirements and Benches as normal.

Switching off is deliberately reversible. A laboratory that drops its early shift over the summer can set Active back to `Y` in September without retyping the times, the colours or the requirements. Deleting the row works too, but you would lose all of that.

If every shift is marked inactive, the roster says so plainly rather than producing an empty rota.

11. Shift Requirements

What each shift needs in order to be safe to run. Every condition is checked **independently** — a senior does not satisfy a transfusion competency, and a competent scientist does not satisfy a coordinator requirement.

COLUMN	NOTES
Shift Code	Must match the Shifts sheet
Days	`All`, `Weekday`, `Weekend`, or a single day. A day-specific row wins over a general one, so you can set a different Saturday minimum without restating everything
Min Staff	Minimum headcount
Min Registered BMS	How many must be registered
Min Senior	How many must be marked Senior
Min Band + Staff At Min Band	Used together: *Min Band* `6` with *Staff At Min Band* `1` means at least one person at Band 6 or above
Min Coordinators	How many shift coordinators
Min Trainers	How many trainers or supervisors
Max Trainees	Cap on trainees. Blank or 0 means no cap

COLUMN	NOTES
Required Competencies	Bench (or discipline) and number, comma separated: `GRP:1, FBC:2, COAG:1`
Required Authorisers	Same format: `XMTCH:1`

How competency requirements are counted

There is an important distinction here, and getting it wrong is the usual cause of an impossible requirement.

- **Sections need distinct people.** A section is a physical place somebody stands. Three sections each needing one competent person need *three different people*, even if one person holds all three competencies.
- **A shift's own competency list does not.** `BT:1, HAEM:1` on a single-handed night shift means *one person competent in both*. Asking for two distinct people on a one-person shift could never be satisfied.

Because sections drive distinct people, section demand also drives staffing: a shift with a headcount minimum of one will still roster three people if three sections need cover.

12. Benches

The laboratory benches that need somebody actually standing at them. **Bench** and **Discipline** are interchangeable terms for the same thing, shown side by side on this sheet: Bench is the short name used on the rota, Discipline is its full name.

COLUMN	NOTES
Bench	The short name used on the rota and in issue headlines, such as `GRP` or `HB'THY`
Discipline	The full name of the same bench, such as *Grouping* or *Haemoglobinopathy*. Competencies and requirements may use either name
Days	`All`, `Weekday` or `Weekend`
Min Staff	How many people during the week
Min Staff Weekend	A reduced weekend service. `0` closes the section at weekends
Shift Codes	Which shifts the section must be staffed during, such as `C, L, W`. Blank means every shift except nights
Requires Authoriser	`Y` means only a result authoriser can cover it
Rotation Interval (days)	Used by the rotation warning

> **Watch the Shift Codes column.** If you leave it blank, every non-night shift is expected to staff that section. A two-person late shift cannot cover four sections, so you will get gaps. List only the shifts the section genuinely runs during.

One person covers one section at a time. Coverage is judged on who is *actually allocated*, not on who happens to be on duty holding the right competency — so a multi-skilled scientist is never counted twice.

13. Leave and Leave Types

Nobody is ever rostered on a day they are recorded as absent.

Leave

COLUMN	NOTES
Staff ID	Must match the Staff sheet
From / To	Both dates are included in the absence
Type	A code from the Leave Types sheet
Credited Hours	Optional. States the hours for this absence explicitly instead of deriving them. Pro-rated if the absence straddles the period edge
Reason	Your own note. Never appears on the staff rota

Why credited hours matter

Without them, somebody who takes a week off still shows a shortfall against their contracted hours, and the rota tries to fill it — giving them a full workload squeezed into fewer days. Crediting the absence closes that gap, so hours lost to leave are not clawed back with extra shifts.

Hours are therefore reported as five figures:

FIGURE	MEANING
Target	Contracted weekly hours scaled to the period
Worked	Hours actually on shifts
Credited absence	Hours credited for time away
Total accounted	Worked + credited. This is what balancing compares
Variance	Total accounted minus target

Leave Types

COLUMN	NOTES
Code / Label	Shown on the rota and in the key
Colour / Font Colour	Hex, displayed as the actual colour
Counts Towards Contracted Hours	`Y` credits the hours, so no make-up shifts are added
Credited Hours Method	`Working pattern` derives them from that person's own days, `Fixed hours per day` uses the column below, `Not credited` credits nothing
Fixed Hours Per Day	Used with the fixed method

The supplied codes are `A/L`, `S/L`, `C/L`, `M/L` and `S/D`, all crediting hours by default. This is a planning setting, not a statement of HR or pay policy — organisations treat sickness and long-term absence differently, so it is configurable rather than assumed.

Because credited hours come from each person's own pattern, a part-timer absent for a week is credited a part-time week.

14. Generating a roster

Choose your completed workbook, then press **Generate draft roster**. Everything happens on your own machine, so nothing is uploaded and the speed depends on your computer rather than a connection.

If the workbook needs correcting you will get a list of everything wrong at once, each naming the sheet, the row and the column heading. Fix them, save, and choose the file again — you do not have to reload the page.

A workbook from an earlier version still works. You will see a note saying it predates the current template, because it will not have the newer optional columns — *Active* on the Shifts sheet and *Initials* on the Staff sheet. Nothing is rejected, and a missing Active column simply means every shift runs, exactly as before.

Advanced settings

Optional. You can override the start and end dates without editing the workbook, which is useful for producing next month from the same file. The *Roster generation ID* is recorded with each draft so a particular rota can be reproduced; you never need to change it by hand.

15. Reading the dashboard

The headline is the **roster status**: *No issues found*, *Review suggested*, or *Attention required*.

The two coverage figures

These are different questions and it matters not to confuse them.

FIGURE	WHAT IT MEASURES
Staffing slot coverage	Whether the positions are occupied. A headcount measure only
Shifts meeting all configured requirements	Whether each shift satisfies *everything* you set: staffing, registration, seniority, band, coordinator, competencies, authorisers, the trainee limit, section coverage, availability and rest

> These routinely differ, and the gap is the useful part. On the challenging example, **100%** of positions are occupied while only **63%** of shifts satisfy everything configured. A full shift is not necessarily a safe one.

The other tiles

TILE	MEANING
Unfilled shifts	Shifts below their minimum headcount
Uncovered benches	Bench slots with nobody competent allocated
Senior cover gaps	Shifts requiring a senior with none
Competency gaps	Missing competencies or authorisers
Rest rule conflicts	Gaps shorter than your configured interval
Outside target hours	People beyond your hours tolerance
Competencies expiring / expired	Against your warning thresholds
Single points of failure	Benches with only one independently competent member of staff
Weekend / night fairness	*Good*, *Review* or *Uneven*

Fairness is a coarse judgement to review, not a score to optimise — there is no single correct fairness formula. Night fairness is measured in *blocks*, because ten three-night blocks shared between eight people cannot come out level, and counting individual nights would call the closest available split unfair.

16. Working through issues

Issues open with counts by severity and the **main causes** ordered by frequency, so you can see where to start rather than reading a list.

SEVERITY	MEANING
Critical	Resolve before publishing: an unfilled shift, a missing competency or authoriser, absent senior cover, somebody rostered while unavailable
Review	Worth a look: materially off contracted hours, uneven distribution, a competency expiring, thin section cover, a single point of failure
Passed	Checked and satisfied. Useful evidence that a rule was applied

Each issue states the date, shift and section, what was **required**, what was **available**, the **impact**, and a suggested review point. For example:

> **Morphology cannot be covered**
> Core. Morphology cannot be staffed on this shift.
> *Required:* 1 independently competent MORPH scientist *Available:* 0
> *What to check:* consider moving a competent scientist onto this shift, changing the deployment plan, or arranging additional cover.

Related warnings with the same underlying cause are combined. A missing morphology scientist produces one issue, not two — one for the competency and another for the bench that consequently could not be staffed.

A short list, not a long one

A roster should leave you with a handful of things to act on, not a wall of red. So findings of one kind are gathered into **one item per cause**: "No senior member of staff on 31 shifts", not thirty-one separate lines. Each item says how many shifts or people it covers and over what dates, and **Show what is behind this** lists every one beneath it. Nothing is hidden; the manager report lists them all too. The counts in the Critical, Review and Passed tabs count items, not findings, and the *Main causes* panel shows how many findings each cause stands for.

17. Roster preview and bench coverage

Staff down the side, dates across the top, shift codes in the cells, coloured as you configured them. The name column stays put while you scroll sideways.

This on-screen preview names people in full. The *exported* rota adds an Initials column, because that is where the section allocation rows use initials — see Exporting.

- **Search staff** by name, role or band
- **Filter by shift** to see who is on nights, for instance
- **Narrow the dates** to focus on a week
- Hover a cell for the shift name, times and any section allocation
- A cell marked `*` was adjusted by hand

- A blank cell means not scheduled

The **Bench coverage** panel below lists each date and shift with the section, how many people were required and allocated, a status of *Covered*, *Review* or *Not covered*, and who is on it. Filter by date or section.

18. Contracted hours

Per person: contracted weekly hours, target for the period, worked, credited absence, total accounted, variance, percentage of target, and a status.

Work is distributed by **hours**, not by counting shifts. Twenty ten-hour nights total 200 hours where twenty seven-and-a-half hour days total 150 — treating those as equivalent would be the wrong answer. Shift counts are reported as a statistic only.

Somebody mildly under target is often just a part-timer whose available days do not line up with demand. Somebody well over is worth investigating: it usually means they are one of very few people who satisfy a constrained requirement.

19. Workforce resilience

How exposed you are if one person is unavailable. Two different risks are reported separately, because they are not the same thing.

LEVEL	QUESTION	EXAMPLE
Workforce	How many people in the whole department can do this?	Only one morphology-competent scientist exists
Shift	How many are on duty for a particular shift?	Only one transfusion authoriser is scheduled on Tuesday

At workforce level, one competent person is **Critical**, two is **Review**, three or more is **Passed**. Shift-level thinness is reported once per shift and bench with a count of affected dates, rather than once per day — a thinly covered bench is one problem to think about, not thirty. It is reported only where a shift has two or more people (a lone worker is a choice you made) and nobody else rostered that day could stand in for the one competent person.

Section rotation is also tracked: if somebody competent in a bench was not allocated to it, that is flagged against your configured interval. It is judged only when the roster is at least as long as that interval, because a shorter roster cannot show a breach of it. No rotation rule is invented; the report simply says how long it has been.

20. Adjusting a roster by hand

The draft is a starting point. To change it, use **Adjust an assignment**:

1. Choose the **date** and **shift**
2. Choose somebody to **take off**, somebody to **put on**, or both
3. Add a reason if you want a record of why
4. Press **Apply change**

Only people who could actually work that shift are offered, each listed with their current competencies. Anybody excluded is counted with the reason — on leave, not their working day, cannot work nights or weekends, or already working elsewhere — so you can see why a name is missing rather than wondering.

Every check then re-runs: staffing, competency and section coverage, hours, fairness, rest rules, resilience, the dashboard and both exports. If your change creates a problem, you will see it immediately.

> **Taking somebody off means they are not rostered anywhere that day.** Without that, the tool would simply put them back when it refilled the shift. To move somebody between shifts, take them off and put them on the other shift in the same change.

Manual changes are marked as such, listed with an **Undo** for each and an **Undo all**, and they survive **Generate alternative roster** — they are never silently overwritten. Undoing everything returns exactly the automatic roster.

Generate alternative roster

Produces a different valid draft from the same information, keeping your workbook, date range and rules. Useful when a rota is technically fine but you would rather the work fell differently. If you have manual changes you will be warned before they are replaced.

Start new roster

Clears the loaded workbook, the draft, the issues, the filters and any manual changes, and returns to the upload state. No page reload, and the tool stays ready to use.

21. Exporting

One draft, two audiences. Both come from the same roster, so exporting one does not change the other or lose your manual adjustments.

Export staff rota

A simplified workbook for circulating to the team: *Roster*, *Section Allocations* and *Notes*. Clean, printable, and deliberately limited to what somebody needs in order to read their own rota.

The Roster sheet carries an **Initials** column beside each name, so the initials used in the section allocation rows can be matched to a person without leaving the sheet.

> **What it withholds, on purpose.** Competency records, individual hours, fairness analysis, workforce vulnerabilities, private working restrictions, leave reasons — and the *type* of any absence. Absence appears as an ordinary non-working day, because printing `S/L` on a departmental rota discloses that somebody was off sick.

Export manager workbook

The full analytical record, in ten sheets:

SHEET	CONTAINS
Instructions	Period, status, and the rules that were applied
Roster	The rota, with an Initials column beside each name, section allocations and per-shift staffing counts
Staff	Who was included, grades, patterns, restrictions
Competencies	The register, with what counts as cover
Shift Requirements	What each shift was required to provide
Bench Allocations	Bench cover day by day, with status
Issues	Everything to review, by severity
Hours Summary	Target, worked, credited, accounted, variance
Fairness Summary	How the unpopular shifts are shared
Competency Expiry	Expired and expiring records

Both are formatted to be usable: freeze panes, filters, sensible column widths, landscape and fit-to-width for printing, and a legend. Status is always carried by a word as well as a colour, so nothing depends on colour alone.

22. How it decides

Useful to know when a result surprises you.

Hard rules it will not break

If a shift cannot be filled without breaking one of these, it is left short and reported:

- Nobody is rostered on a day they are recorded as absent
- One shift per person per day
- Nobody is rostered outside their working pattern or available hours
- No shift the day after a night, and the configured rest interval is honoured in both directions
- Maximum consecutive days and nights
- Nights only for staff marked as night-capable; weekends likewise
- Personal caps on nights, weekends, days per week and hours

Order of work

1. **Nights first.** They are the most constrained and they generate the recovery days everything else has to work around
2. **Then day shifts, hardest first** — the shift with the fewest eligible people goes first
3. **Within a shift, the rarest capability first**, so a scarce competency is not spent on a slot somebody else could have filled
4. **Then top up** to the minimum headcount
5. **Then allocate sections**, one person to one section

Who gets offered work

Among eligible people, whoever is furthest below their contracted target goes first. For weekend days, whoever has done fewest weekends leads instead — ordering by hours alone concentrated weekends on full-time staff, because a larger target means a larger shortfall. Nights are shared the same way when *Share nights evenly* is on.

Fairness comparisons are only ever made within the group eligible for that work, so somebody who cannot work nights is never counted as having an unfairly light night load.

23. If something goes wrong

The workbook is validated before any rota is produced, and every problem is reported at once so you can fix them in one pass. Each names the sheet, the row and the column heading rather than an internal error.

MESSAGE	WHAT TO DO
That file could not be opened as an Excel workbook	It may be a different file type, an old `.xls`, or damaged. Download a fresh workbook
This workbook was made with an earlier version	It lacks competency, hours or requirement information that cannot be safely guessed. Download the latest blank workbook and copy your staff into it
The Staff sheet is missing the '…' column	Add the heading, or start from a fresh workbook
Staff ID '…' is used twice	IDs must be unique; the other sheets refer to them

MESSAGE	WHAT TO DO
'...' is not a competency status	Use one of the seven listed statuses. The dropdown offers them
... is marked as not working on any day of the week	Mark at least one day, or clear all seven to mean fully flexible
... has a latest finish at or before their earliest start	No shift could ever fit; correct one of the two times
Shift '...' starts and finishes at the same time	Check the start and finish times
Shift '...' has no entry on the Shift Requirements sheet	Add a row saying how many people it needs, or no staff will be rostered onto it
Nobody holds a '...' competency	A bench can never be covered. Check the bench or discipline written on the Competencies sheet matches the Benches sheet
Could not write ... the file is open in another program	Close it in Excel and try again

The rota looks wrong rather than broken

- **Somebody is never rostered.** Check their Mon–Sun availability, night and weekend eligibility, and any personal caps
- **Nobody is rostered at weekends.** Their availability pattern probably excludes Saturday and Sunday
- **A bench is never covered.** Check the bench or discipline written on the Competencies sheet matches, that somebody is *Competent* rather than *In Training*, that the competency has not expired, and whether the section requires an authoriser
- **Everybody is over or under target hours.** Your total shift demand does not match your total contracted hours. Adjust the requirements or the hours
- **Too many people are under target.** Often a sign that a requirement routes work to a small group, leaving others idle

24. What it does not do

- **It does not make staffing decisions.** Every rota is a draft for managerial review
- **It is not artificial intelligence.** It is a rules engine. Where it cannot fill a shift without breaking a rule you set, it leaves the shift short and says so
- **It does not determine legal or regulatory compliance.** Rest intervals and other limits are rules you configure
- **It does not replace** local workforce policies, HR systems, payroll, ESR or professional judgement
- There are no user accounts, no external database, no cloud storage and no integrations

Not yet supported

- **Bank holidays** are not modelled as a distinct category
- **Section rotation** is reported but is not yet an input to scheduling
- **Part-shift sections** — morphology 09:00–13:00 then coagulation 13:00–17:00 — are not yet available; sections are covered for a whole shift
- The engine produces a workable draft reliably but does not prove it is the best possible one. A human-built rota may sometimes be better

---

Lab Shift Roster is a product of Optymum SS (UK) Ltd.